

NO	ANSWERS	MARKS
1 a) i	$k = 20\pi$ $k = 20\pi = \frac{2\pi}{\lambda}$ $\lambda = 0.1 \text{ cm}$	G1 JU1
ii	$w = \frac{4\pi}{2} = 2\pi$ $w = 2\pi f = 2\pi$ $f = 1 \text{ Hz}$	G1 JU1
iii	$v = f\lambda$ $v = 0.01(1)$ $v = 0.01 \text{ cm s}^{-1}$	G1 JU1
iv	$v = \frac{dy}{dt} = 4\pi \cos(2\pi t + 20\pi x)$ $v = 4\pi \cos(2\pi (0.5) + 20\pi (300))$ $v = -4\pi \text{ cm s}^{-1}$	K1 G1 JU1
1 b) i	$w = 5\pi = \frac{2\pi}{T}$ $T = 0.4 \text{ s}$ $k = 3\pi = \frac{2\pi}{\lambda}$ $\lambda = 0.67 \text{ cm}$	G1 JU1 G1 JU1
ii	<i>distance bet node and antinode</i> $= \frac{\lambda}{4}$ $= 0.168 \text{ cm}$	K1 JU1
iii	<i>amplitude</i> $= 15 \cos(3\pi x)$ <i>amplitude</i> $= 15 \cos(3\pi 0.5)$ <i>amplitude</i> $= 0 \text{ cm}$	G1 JU1
TOTAL		17 marks

NO	ANSWER	MARKS
2(a)(i)	$It = ne$ $(0.6)(2 \times 60) = n(1.60 \times 10^{-19})$ $n = 4.50 \times 10^{20} \text{ electrons}$	G1 JU1
2(a)(ii)	$V = IR$ $15 = (0.6)R$ $R = 25\Omega$	GJU1
2(a)(iii)	$\rho = \frac{RA}{l}$ $\rho = \frac{(25)(\pi(0.3 \times 10^{-2})^2)}{2.80}$ $\rho = 2.52 \times 10^{-4} \Omega m$	G1 JU1
2(b)(i)	$V = IR$ $12 = 0.2R$ $R = 60\Omega$ $R_{eff} = R_1 + R_2$ $60 = 2R$ $R = 30\Omega$ OR $V = IR$ $6 = 0.2R$ $R = 30\Omega$	G1 JU1 OR G1 JU1
2(b)(ii)	$I = 0.5 - 0.2 = 0.3A$ $V = IR$ $12 = 0.3R$ $R_{eff} = 40\Omega$ $R_{eff} = R_1 + R_2$ $40 = 28 + R$ $R = 12\Omega$	K1 G1 GJU1
2(b)(iii)	$V = IR$ $V = (0.3)(28)$ $V = 8.4V$	G1 JU1
	TOTAL	12

	$V_2 = \frac{Q_2}{C_2} = \frac{300 \mu}{22 \mu} = 13.6 V$	
3 (b)	$Q = 30\% Q_0 = 0.3 Q_0$ $Q = Q_0 \left(1 - e^{-\frac{t}{RC}} \right)$ $0.3 Q_0 = Q_0 \left(1 - e^{-\frac{t}{(100 \times 10^3)(58 \times 10^{-6})}} \right)$ $t = 2.07 s$	G1 JU1
TOTAL		9 MARKS

NO	SOLUTION	MARKS
4(a)		
i.	$B_{solenoid} = \frac{\mu_0 NI}{L}$ $(2.3 \times 10^{-3}) = \frac{(4\pi \times 10^{-7})(N)(16)}{(2)}$	G1
ii.	$N = 228.785$ $L = 2\pi \left(\frac{d}{2}\right)(N)$ $= 2\pi \left(\frac{3.8 \times 10^{-2}}{2}\right)(N)$ $= 27.31 \text{ m}$	JU1 GJU1
4(b)	$B_p = \frac{\mu_0 I_p}{2\pi r_p}$ $= \frac{(4\pi \times 10^{-7})(40)}{2\pi(10 \times 10^{-2})}$ $= 8.00 \times 10^{-5} T$ <i>Into the page</i>	G1 JU1 J1

4(c)		
i.	$\Delta A = A_f - A_i$ $= \frac{(\pi)(5.3 \times 10^{-2})^2}{4} - \frac{(\pi)(4 \times 10^{-2})^2}{4}$ $= 9.495 \times 10^{-4} m$	K1 GJU1
ii.	$\varepsilon_{ind} = -NB \frac{dA}{dt}$ $= -(10)(1.2) \frac{(9.495 \times 10^{-4})}{(0.25)}$ $= 0.0456 \text{ V}$	G1 JU1
iii.	$\varepsilon_{ind} = I_{ind} R$ $0.0456 = I_{ind} (2.4)$ $I_{ind} = 0.019 \text{ A}$	G1 JU1
iv.	$Q = I_{ind} t$ $= (0.019)(0.25)$ $= 4.748 \times 10^{-3} \text{ C}$	G1 JU1
	TOTAL MARKAH	14 MARKS

NO.	SOLUTION	MARKS
5(a)(i)	$u + v = 30 \dots \dots \dots (1)$ $m = -\frac{v}{u}$ $2 = -\frac{v}{u}$ $v = -2u$ (substitute into equation 1) Equation 1 becomes $u + 2u = 30$ Therefore, $u = 10 \text{ cm}$ Also, $v = -2u = -20 \text{ cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{f} = \frac{1}{10} - \frac{1}{20}$ $f = 20 \text{ cm}$	K1 G1 G1 JU1
5(a)(ii)	Concave mirror	J1
5(b)	$m = -\frac{v}{u}$ $3 = -\frac{v}{u}$ $v = -3u$ $r = 2f$ $50 = 2f$ $f = 25 \text{ cm}$	G1 G1

TOTAL		14
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(a)

$$\begin{aligned}\Delta y &= \frac{\lambda D}{d} \\ &= \frac{(622 \times 10^{-9})(0.5)}{0.3 \times 10^{-3}} \quad \text{G1} \\ &= 1.04 \text{ mm}\end{aligned}$$

JU1

- (ii) As Δy is directly proportional to wavelength J1
 the separation between consecutive fringes will be decreased J1
 when shorter wavelength is used

(b) (i)

$$\begin{aligned}y &= \frac{n\lambda D}{d} \\ n &= 1 \quad \text{K1} \\ y_1 &= \frac{(1)\lambda(2)}{4.66 \times 10^{-3}}\end{aligned}$$

$$\lambda = 575 \text{ nm} \quad \text{GJU1}$$

$$\begin{aligned}\text{(ii)} \quad y &= \frac{n\lambda D}{d} \\ n &= 8 \quad \text{K1}\end{aligned}$$

$$y_{8\text{bright}} = \frac{(8)(575 \times 10^{-9})(2)}{4.66 \times 10^{-3}} = 1.97 \text{ mm} \quad \text{G1}$$

$$\begin{aligned}n &= 2 \quad \text{K1} \\ y_{2\text{dark}} &= \frac{\left(2 + \frac{1}{2}\right)(575 \times 10^{-9})(2)}{4.66 \times 10^{-3}} = 0.62 \text{ mm} \quad \text{G1}\end{aligned}$$

$$\text{Distance} = y_{8\text{bright}} - y_{2\text{dark}} = 1.35 \text{ mm} \quad \text{GJU1}$$

(c)

$$y_{1blue} = \frac{n\lambda D}{d} = \frac{(1)(400 \times 10^{-9})(60 \times 10^{-2})}{0.08 \times 10^{-3}} = 3 \text{ mm} \quad \text{G1}$$

$$y_{2yellow} = \frac{n\lambda D}{d} = \frac{(2)(600 \times 10^{-9})(60 \times 10^{-2})}{0.08 \times 10^{-3}} = 9 \text{ mm} \quad \text{G1}$$

$$y_{2yellow} - y_{1blue} = 9 - 3$$

$$= 6 \text{ mm} \quad \text{GJU1}$$